1 A car accelerates uniformly in a straight line from rest at time $t = 0$. At $t = 3.2\,s$, the speed of the car is $13.0\,m/s$.

(a) (i) Calculate the acceleration of the car.

acceleration = ... [2]

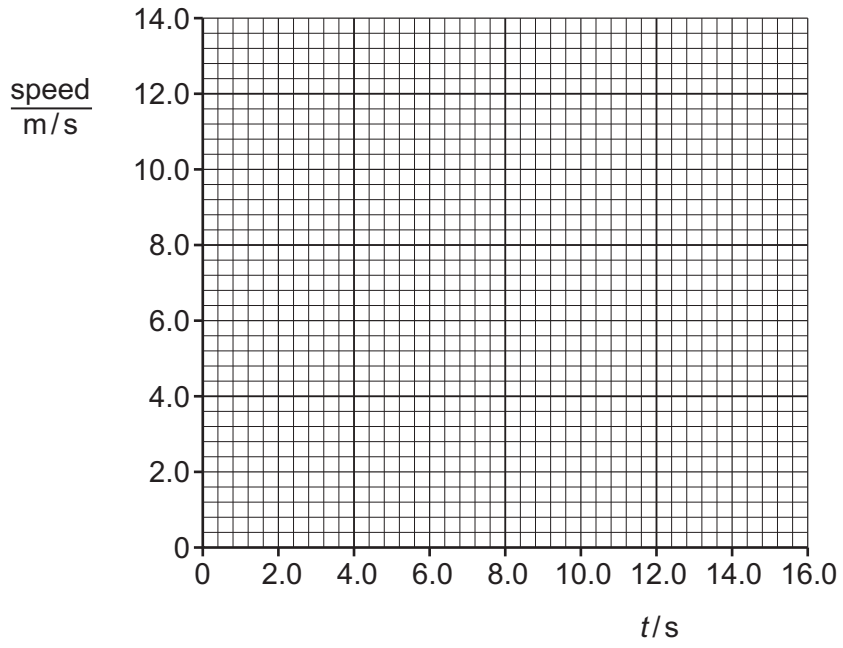
(ii) Explain in words what is meant by the term acceleration.

.....

..... [1]

(b) The car travels at $13.0\,m/s$ from $t = 3.2\,s$ to $t = 12.0\,s$.

(i) Plot the speed–time graph for the car from $t = 0$ to $t = 12.0\,s$.



[2]

(ii) Determine the distance travelled by the car between $t = 0$ and $t = 3.2\,s$.

distance = ... [2]

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(c) The car decelerates from 13.0 m / s to 0 m / s at a constant deceleration. The mass of the car is 1350 kg. The car travels 13 m in 2.0 s as it decelerates.

Show that the work done by the car as it decelerates is approximately 1.1×10^5 J.

[4]

(d) On another day, the car in **(c)** travels a longer distance while it decelerates from 13.0 m / s to 0 m / s. The deceleration is constant.

Suggest and explain what causes the stopping distance to increase.

suggestion .....

.....

explanation .....

.....

[2]

[Total: 13]